ABREHAM DESSU

Full Stack Software Engineer

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SUMMARY

Full Stack Software Engineer with **4+ years of experience** shipping complete products — from PostgreSQL schema and event-driven microservices through to the React and Next.js interfaces users touch. Scaled a Kafka-based microservices platform to **10,000+ users**, engineered payment infrastructure processing **5,000+ daily transactions at a 96% success rate**, and improved system reliability **30%** while reducing downtime **40%**. Fluent across Node.js, NestJS, TypeScript, React, Next.js, and Angular, with production deployment experience on Google Cloud Platform and AWS. **AWS Certified Solutions Architect – Associate.** Works fluently with AI-assisted development tooling (Claude Code, Cursor, GitHub Copilot, Codex) to accelerate delivery without sacrificing code quality or review discipline.

SKILLS

Programming Languages
JavaScript • TypeScript • Python • Go • SQL • Bash

Frontend Technologies
React • Next.js • Angular • Tailwind CSS • HTML5 • CSS3 • Responsive Design • Component-Based Architecture • Component Stories (Storybook) • SPA • Server-Side Rendering

Backend Technologies
Node.js • NestJS • Express.js • Django • Flask • REST APIs • GraphQL • Microservices • Distributed Systems • Event-Driven Architecture • Apache Kafka • Webhooks • Third-Party API Integration • Fault Tolerance

Databases
PostgreSQL • Redis • MySQL • MongoDB • DynamoDB • Prisma ORM • Meilisearch • Database Design • Query Optimization • Schema Migrations

Cloud Platforms
GCP (Cloud Run, Cloud Build) • AWS • Microsoft Azure (AKS) • Cloud Architecture • Scalability

DevOps and Infrastructure
Docker • Kubernetes • Terraform • GitHub Actions • Azure DevOps • CI/CD • Prometheus • Grafana • Production Monitoring • Incident Response

Testing and Quality
Jest • Vitest • Supertest • Storybook • Cypress • Playwright • Unit & Integration Testing • API Testing • E2E Testing • TDD • Code Review • Debugging

Security and Authentication
Firebase Auth • OAuth 2.0 • JWT • Role-Based Access Control (RBAC)

AI-Assisted Development
Claude Code • Cursor • GitHub Copilot • Codex • Prompt Engineering • Amazon PartyRock

Tools and Practices
Git • GitHub • Postman • Jira • Agile / Scrum • SDLC • System Design • Data Structures & Algorithms

EXPERIENCE

Full Stack Engineer — Jun 2025 – Present | Addis Ababa, Ethiopia
ETM Software

- Improved system reliability **30%** by re-architecting the backend service layer and implementing production monitoring, alerting, and automated CI/CD deployment pipelines on Google Cloud Platform.
- Engineered a fault-tolerant integration keeping **5,000+ HubSpot company and contact records** synchronized with near-zero data drift, using retry logic, deduplication, and reconciliation to survive partial API failures and rate limiting.
- Eliminated manual data entry across **2,000+ hierarchical entities** by automating multi-level synchronization workflows, removing a recurring source of data corruption in business-critical records.
- Reduced feature delivery time **25%** by designing **5+ reusable service modules** in NestJS and Prisma ORM, establishing shared data-access and validation patterns across the codebase.
- Secured business-critical operations with **zero authentication-related incidents to date**, implementing Firebase Authentication with JWT access control and role-based permissions.
- Designed and deployed GraphQL and RESTful APIs on NestJS and PostgreSQL, collapsing multi-call data fetching into single round trips and enabling client-side workflow automation for internal teams.

Full Stack Developer — Sep 2023 – Jun 2025 | Addis Ababa, Ethiopia
Ablaze Labs

- Scaled payment infrastructure to **5,000+ daily transactions at a 96% success rate** by integrating Chapa and SantimPay gateways with idempotent request handling, webhook signature verification, and automated failure recovery.
- Reduced production downtime **40%** and improved performance **20%** through query profiling, PostgreSQL index tuning, and disciplined production incident triage and root-cause analysis.
- Scaled an event-driven microservices gaming platform to **10,000+ users**, building both the NestJS/Kafka backend services and the React interfaces for user management, game management, referrals, and payments.
- Led end-to-end delivery of a national content management system for Ethiopia's **Ministry of Water and Energy**, architecting backend APIs and building the frontend content management modules across **12 interconnected subprojects**.

- Replaced manual, document-driven government reporting with automated APIs and CMS interfaces, reducing report turnaround from **3 days to under 2 hours** across multiple ministry projects.
- Increased user acquisition **35%** on a job board platform by shipping full stack employer and applicant workflows, subscription billing, and a referral-based growth engine.
- Delivered Problem-Based Learning functionality for a medical e-learning platform, building the collaborative case-session interface and supporting services used by **500+ students and instructors**.

Junior Software Developer Oct 2022 – Aug 2023 | Addis Ababa, Ethiopia
Hirundo Security Solution

- Improved deployment efficiency **22%** by automating server configuration and provisioning across **10+ machines** with repeatable, scripted Bash workflows.
- Increased IT support efficiency **15%** by developing internal backend systems and administrative web applications that automated the team's highest-volume manual processes.
- Consolidated fragmented operational workflows into a unified web-based toolset, measurably reducing manual workload for business teams.
- Strengthened stability of business-critical systems through unit and integration testing, debugging, and resolution of production defects before user impact.

PROJECTS

Microservices Gaming Platform NestJS · Kafka · PostgreSQL · React
- Event-driven backend serving **10,000+ users** across independently scalable services for user management, game management, referrals, and payments, paired with a React frontend for player onboarding, game browsing, and wallet management.

Water & Energy CMS — Ministry of Water and Energy NestJS · PostgreSQL · React · Docker
- Nationwide platform spanning **12 interconnected subprojects**; designed the shared backend API layer and built the content management interfaces that standardized reporting for government stakeholders.

HubSpot Enterprise Integration Service NestJS · Prisma ORM · PostgreSQL · GCP
- Bidirectional synchronization keeping **5,000+ CRM records** and **2,000+ hierarchical entities** consistent between HubSpot and internal PostgreSQL systems under real-world API rate limits and partial failures.

Job Board & Talent Marketplace Next.js · NestJS · Meilisearch
- End-to-end employer and applicant experiences — job posting, application tracking, subscription billing, and referral-driven acquisition — with responsive server-rendered Next.js interfaces and sub-second faceted search.

EDUCATION

A2SV (African to Silicon Valley) Jan 2025 – Jan 2026 | Addis Ababa, Ethiopia
Software Engineering Trainee — Data Structures and Algorithms

Adigrat University Nov 2016 – Jul 2022 | Adigrat City, Ethiopia
Bachelor of Science in Electrical and Computer Engineering

CERTIFICATIONS

AWS Certified Solutions Architect – Associate
— Amazon Web Services

AWS Certified Cloud Practitioner
— Amazon Web Services

Microsoft Azure Fundamentals (AZ-900)
— Microsoft

AWS Scholars: Analyze Data using AI with PartyRock
— Udacity, Mar 2026

AWS Scholars: Build Your First AI Productivity App
— Udacity, Mar 2026

Professional Foundations
— ALX Africa

AI Career Essentials
— ALX Africa

IELTS Academic — Band 6.5 (CEFR B2)
— British Council, Oct 2025